DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 3: Perform Data Cleaning and Missing Value Treatment Using WEKA

Aim: To identify missing values and perform data cleaning operations on a dataset using WEKA.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the importance of data quality.
2. Identify missing values in datasets.
3. Detect duplicate and inconsistent records.
4. Apply data cleaning techniques using WEKA.
5. Use filters for handling missing values.
6. Compare datasets before and after cleaning.
7. Prepare data for further preprocessing and mining.

Importance of this Experiment:

Data collected from real-world applications is rarely complete and error-free. Before applying any data mining algorithm, the dataset must be cleaned and prepared. Poor-quality data can:

- Reduce prediction accuracy.
- Generate incorrect patterns.
- Affect clustering and classification results.
- Produce misleading analytical outcomes.

Therefore, data cleaning is considered one of the most important stages of the data mining process.

Expected Outcome: After completing this experiment, Students will be able to:

1. Detect missing values in datasets.
2. Apply data cleaning techniques using WEKA.
3. Handle missing values using appropriate filters.
4. Improve the quality of datasets.
5. Prepare datasets for preprocessing, classification, clustering, and association rule mining experiments.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Sample Dataset (student_missing.arff)

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1: Installation and Exploration of WEKA Tool and Available Datasets
- Experiment No. 2: Create and Analyze ARFF Files Using WEKA
- Basic WEKA Environment
- ARFF File Structure
- Attributes and Instances
- Data Mining Fundamentals

Theory:

1. Data Preprocessing

Data preprocessing refers to the process of preparing raw data before applying data mining algorithms. The main objectives of preprocessing are:

- Improving data quality
- Increasing accuracy
- Reducing inconsistencies
- Improving analytical efficiency

Data cleaning is one of the most important activities in preprocessing.

2. Data Quality Problems

Real-world datasets often contain various quality issues:

1. Missing Values

   Information for some attributes is unavailable.

   Example: Attendance = ?

2. Duplicate Records

   The same record appears more than once.

   Example:

   80,5,Pass

   80,5,Pass

3. Noisy Data

   Data containing errors or random variations.

   Example: Attendance = 800 instead of: Attendance = 80

4. Inconsistent Data

   Data values violate expected rules.

   Example: Age = -5

5. Outliers

   Values that differ significantly from normal observations.

   Example: Attendance = 500 in a dataset where attendance should be between 0 and 100.

3. Data Cleaning

Data Cleaning is the process of detecting, correcting, or removing inaccurate, incomplete, or inconsistent data. Benefits of data cleaning include:

- Improved dataset quality
- Better mining results
- Higher prediction accuracy
- Reduced processing errors

4. Missing Values

A missing value occurs when information for a particular attribute is unavailable.

Example:

Child ID	Age	Weight	BMI
101	4	?	13.2

In the above record, the Weight value is missing.

5. Why Missing Values Are a Problem

Missing values can:

- Reduce dataset quality.
- Affect classification accuracy.
- Distort clustering results.
- Create incorrect analytical conclusions.
- Lead to unreliable recommendations.

Therefore, they should be detected and handled before mining.

6. Techniques for Handling Missing Values

1. Remove the Record
   - Delete records containing missing values.
2. Manual Replacement
   - Enter values manually using expert knowledge.
3. Statistical Replacement
   - Replace missing values using:
     - Mean
     - Median
     - Mode
4. Automatic Replacement
   - Use data mining tools such as WEKA to replace missing values automatically. In this experiment, WEKA's **ReplaceMissingValues** filter will be used.

Industry Insight: In practical Machine Learning and Data Mining projects, data cleaning and preparation may consume 60-80% of the total project effort, whereas model building often requires comparatively less effort.

Mini Real-World Example:

Food Recommendation System

Consider the following dataset:

Child ID	Age	Weight	Nutritional Status
101	4	?	Underweight
102	5	15	Normal

Question: Can a food recommendation system generate accurate recommendations if the child's weight is missing?

Answer: No.

The missing value must first be identified and handled before recommendations can be generated. This demonstrates the importance of data cleaning in real-world applications.

Consequences of Poor Data Quality:

If missing values are ignored:

- Nutritional analysis may become inaccurate.
- Prediction accuracy may decrease.
- Recommendation systems may generate incorrect outputs.
- Analytical results may become unreliable.

Therefore, maintaining data quality is essential for effective decision-making.

Dataset Used in This Experiment:

Filename: student_missing.arff

Dataset Contents:

```
@relation student
@attribute attendance numeric
@attribute studyhours numeric
@attribute result {Pass,Fail}
@data
80,5,Pass
?,2,Fail
90,6,Pass
70,?,Pass
50,1,Fail
```

Dataset Information:

- Relation Name: student
- Attributes: 3
- Instances: 5
- Missing Values: 2
- Numeric Attributes: 2
- Nominal Attributes: 1

Representation of Missing Values:

In ARFF files, missing values are represented using: ?

Before Cleaning vs After Cleaning:

Before Cleaning

Attendance	Study Hours	Result
80	5	Pass
?	2	Fail
90	6	Pass
70	?	Pass
50	1	Fail

After Cleaning

Attendance	Study Hours	Result
80	5	Pass
Replaced Value	2	Fail
90	6	Pass
70	Replaced Value	Pass
50	1	Fail

Note: The replacement values are automatically determined by the WEKA filter. For numeric attributes, WEKA generally replaces missing values using the mean value of the attribute, while for nominal attributes it typically uses the most frequent value.

Procedure:

Activity 1: Load Dataset Having Missing Values

Steps:

1. Open WEKA.
2. Select Explorer.
3. Open the Preprocess tab.
4. Click Open File
5. Select student_missing.arff.
6. Click Open

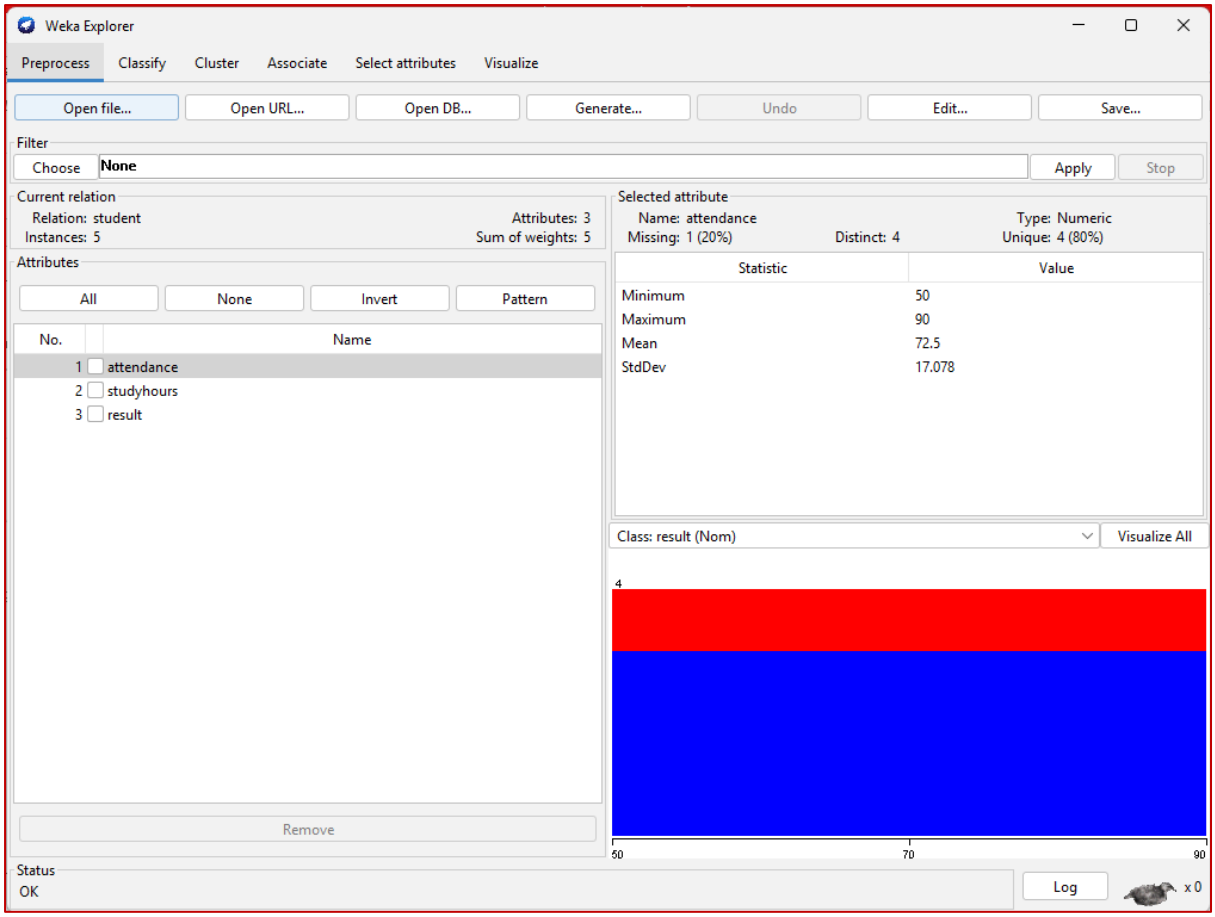


Figure 3.1: Dataset Containing Missing Values

Activity 2: Identify Missing Values

Steps:

1. Select each attribute.
2. Observe the attribute statistics.
3. Identify attributes containing missing values.
4. Record the number of missing values.

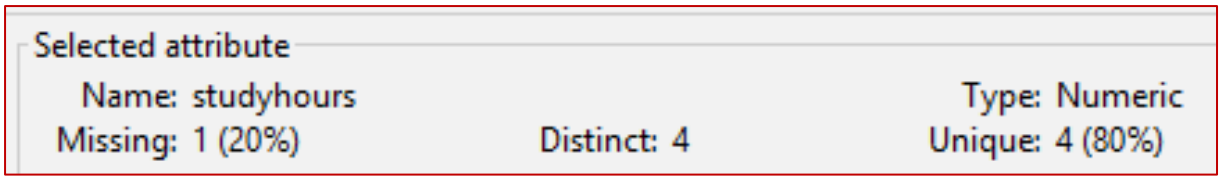


Figure 3.2: Attribute Statistics Showing Missing Values

Activity 3: Apply ReplaceMissingValues Filter

Steps:

1. Navigate to the Preprocess tab.
2. Click: Choose under the Filter section.

3. Select Unsupervised –> Attribute –> ReplaceMissingValues
4. Click Apply
5. Wait for WEKA to process the dataset.

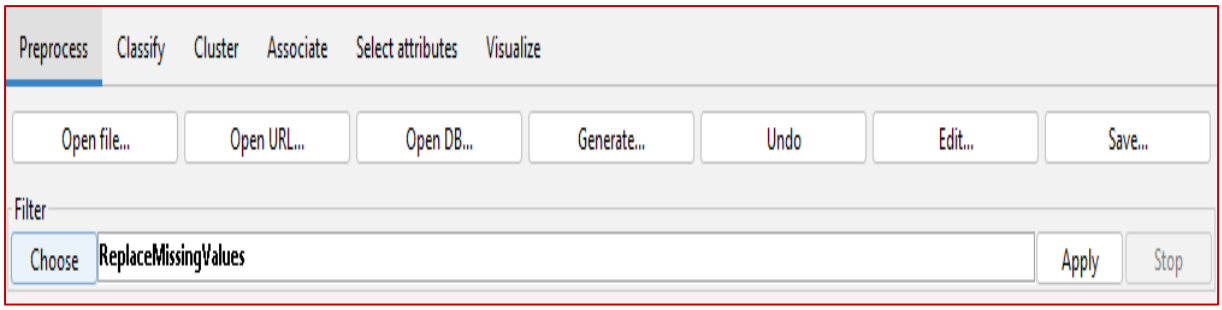


Figure 3.3: Applying ReplaceMissingValues Filter

Activity 4: Verify Cleaned Dataset

Steps:

1. Observe the attribute information again.
2. Check that missing values have been replaced.
3. Compare the dataset before and after cleaning.
4. Record the observations.

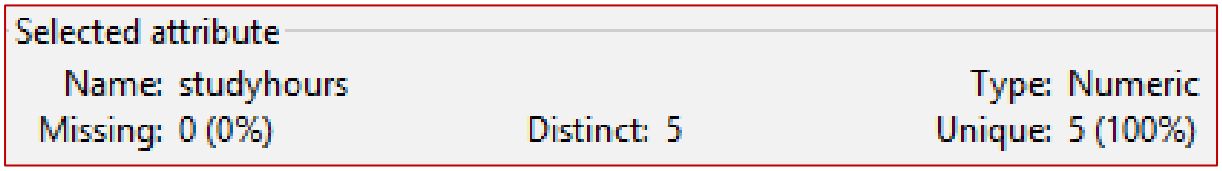


Figure 3.4: Dataset After Missing Value Treatment

Common Mistakes While Handling Missing Values

1. Ignoring Missing Values
   - Students often directly apply mining algorithms without cleaning the dataset.
   - Impact: Poor analytical results.
2. Assuming Missing Values Are Zero
   - Missing values (?) do not necessarily mean zero.
   - Impact: Incorrect analysis.
3. Applying Classification Before Cleaning
   - Algorithms should be applied after preprocessing.
   - Impact: Reduced prediction accuracy.
4. Not Verifying Results
   - Students may apply a filter but fail to verify whether missing values have been removed.
   - Impact: Dataset may still contain data quality issues.
5. Saving Results Without Comparison

- Students should compare datasets before and after cleaning.
- Impact: Learning objectives are not achieved completely.

Common Discussion Questions

1. Can classification be performed on incomplete data?
2. Are missing values always caused by human errors?
3. What could happen if missing values are ignored?
4. Should records containing missing values always be deleted?
5. Why is data cleaning considered an important preprocessing step?

Student Activity:

Create an ARFF dataset satisfying the following requirements:

- Minimum 5 records
- Minimum 3 attributes
- At least 2 missing values

Apply: ReplaceMissingValues filter and record the results.

Record:

- Missing values before cleaning
- Missing values after cleaning
- Replaced attributes

Observation Table:

PARAMETER	OBSERVATION
Dataset Name	
Dataset File Name	
Number of Attributes	
Number of Instances	
Missing Values Before Cleaning	
Missing Attribute(s)	
Missing Values After Cleaning	
Replaced Attribute(s)	
Filter Applied	

Result: The dataset containing missing values was successfully loaded and analyzed using WEKA. Missing values were identified and treated using the ReplaceMissingValues filter. The importance of data cleaning and preprocessing in improving data quality for data mining applications was studied successfully.

Viva Questions:

1. What is data preprocessing?
2. Define data cleaning.
3. What are missing values?
4. Why are missing values problematic?
5. What is the purpose of the ReplaceMissingValues filter?
6. Which WEKA tab is used for data cleaning?
7. What are the different types of data quality problems?
8. Can missing values affect classification results?
9. Can missing values affect clustering results?
10. How can missing values be handled?
11. What is noisy data?
12. What are duplicate records?
13. What is an outlier?
14. Why should data be cleaned before mining?
15. What is the importance of data quality?

Assignment Questions:

1. Explain the importance of data cleaning in data mining.
2. Describe different methods used for handling missing values.
3. Create a dataset containing missing values and clean it using WEKA.
4. Differentiate between raw data and cleaned data.
5. Explain the impact of poor-quality data on analytical results.
6. What are the major data quality issues found in real-world datasets?
7. Explain the ReplaceMissingValues filter with a suitable example.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

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